Hybrid-Electric Propulsion Optimization for a Fixed-Wing UAV

Problem Statement

Indian Institute of Technology Indore
In collaboration with
Hindustan Aeronautics Limited (HAL)

1. Background

Hybrid-electric propulsion is an emerging aerospace technology that combines conventional fuel-powered gas turbine engines, Wankel engines, or Internal Combustion (IC) engines with electric propulsion systems to improve endurance, efficiency, and operational flexibility.

This challenge focuses on the conceptual design and optimization of a hybrid-electric propulsion architecture for a fixed-wing UAV mission, preferably employing a turboshaft engine with a rated power of approximately 60 kW.

2. Objective

Design and optimize a hybrid-electric propulsion system for the UAV configuration defined below.

Teams must develop a methodology/framework to:

- Size the propulsion system (preferably a gas turbine engine such as a turbojet or turboshaft).
- Estimate mission feasibility.
- Optimize the overall system performance.

3. Baseline UAV Specifications

Parameter	Value
UAV Type	Fixed-Wing UAV
Maximum Take-Off Weight (MTOW)	~1000 kg
Payload Capacity	~200 kg
Cruise Speed	~250 km/h
Cruise Altitude	3–10 km

4. Design Variables

Participants may optimize:

- Gas Turbine Engine (Turboshaft Engine) size
- Generator architecture (AC/DC)
- Battery chemistry and capacity
- Number of propulsion motors
- Power-sharing strategy between thermal and electric systems

5. Mission Profile

The UAV mission consists of:

- Take-off
- Climb
- Cruise
- Loiter
- Descent and Landing

The primary objective is to maximize endurance while satisfying mission and system constraints.

6. Constraints

The final design must:

- Remain within MTOW limits.
- Carry the specified payload.

- Complete all mission phases.
- Satisfy propulsion power requirements.
- Maintain safe battery operating limits.

7. Assumptions

Participants may use reasonable engineering assumptions and publicly available data sources for:

- Propulsion efficiencies
- Battery characteristics
- Fuel consumption
- Aerodynamic estimates
- Component sizing

All assumptions must be clearly documented and justified.

8. Deliverables

The final submission should include the following.

8.1 Technical Report

- Problem formulation and design methodology
- Propulsion system architecture
- Component sizing methodology
- Mission analysis and power management strategy
- Optimization approach and objective functions
- Design assumptions and engineering justification
- Performance evaluation and trade-off analysis

8.2 Source Code

Complete implementation of the proposed optimization framework, including simulation models, sizing algorithms, optimization routines, and any supporting scripts used for analysis.

8.3 Simulation Dashboard

Interactive visualization demonstrating

- Mission profile and flight phases
- Power distribution between thermal and electric propulsion systems
- Battery State of Charge (SoC)
- Fuel consumption throughout the mission
- Engine and motor operating conditions
- System efficiency
- Endurance estimation
- Optimization results and design trade-offs

8.4 Presentation

A concise summary explaining

- Propulsion architecture selection
- Optimization strategy
- Engineering rationale
- Key design trade-offs
- Final optimized configuration
- Mission performance and endurance improvements

9. Evaluation Criteria

Criteria	Weightage (%)
Mission Feasibility	20
Optimization Quality	25
Engineering Justification	20
Innovation	15
Endurance Improvement	10
Presentation & Visualization	10

10. Notes

1. Detailed CFD, structural analysis, and thermal simulations are not required.
2. The focus shall be on system-level design space exploration and propulsion architecture optimization.

Physics-Informed Digital Twin for Real-Time Four-Stage Turbojet Health Monitoring

Problem Statement

Indian Institute of Technology Indore
In collaboration with
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1. Background

Modern aerospace propulsion systems operate under demanding thermal, mechanical, and aerodynamic conditions. Throughout their operational life, engine components experience gradual degradation due to phenomena such as compressor fouling, turbine erosion, and combustor efficiency loss. These degradation mechanisms affect engine performance, fuel efficiency, thrust generation, and operational reliability.

Traditional engine monitoring approaches rely on direct sensor measurements and periodic inspections. However, many critical engine states and component health indicators cannot be measured directly during operation. Consequently, aerospace industries are increasingly adopting Digital Twin technologies that combine sensor data, engineering knowledge, and intelligent models to estimate hidden engine states and predict future performance.

High-fidelity engine simulations provide valuable insight into engine behavior but are computationally expensive and unsuitable for real-time applications. Surrogate models offer a practical alternative by approximating complex engine behavior with significantly lower computational cost while enabling continuous health assessment and monitoring.

2. Objective

Develop a Physics-Informed Digital Twin capable of reconstructing the operational and health state of a single-spool four-stage turbojet engine using limited sensor measurements.

The proposed framework should estimate hidden subsystem health indicators, predict engine performance, and maintain a continuously updated virtual representation of the engine throughout its operational life.

Participants are expected to combine engineering principles, data-driven methods, and surrogate modeling techniques to develop an interpretable and computationally efficient solution.

3. Dataset Description

3.1 Official Dataset Repository

The complete dataset, along with supporting files and documentation, can be accessed using the following Google Drive repository:

Official Dataset

A synthetic but physics-based dataset generated from a four-stage single-spool turbojet engine model is provided.

The dataset includes multiple virtual engines operating under varying flight conditions and progressive degradation scenarios.

3.2 Available Measurements

Parameter	Unit
Engine ID	–
Cycle	–
Altitude	m
Mach Number	–
Ambient Temperature (T_{amb})	K
Ambient Pressure (P_{amb})	Pa
Shaft Speed (RPM)	rev/min
Fuel Flow Rate	kg/s
Compressor Exit Pressure (P_2)	Pa
Compressor Exit Temperature (T_2)	K
Combustor Exit Pressure (P_3)	Pa
Turbine Inlet Temperature (T_3)	K
Turbine Exit Pressure (P_4)	Pa
Turbine Exit Temperature (T_4)	K

The dataset represents engine operation over multiple degradation cycles and varying flight conditions.

4. Challenge Tasks

4.1 1. Engine Digital Twin Construction

Create a digital representation of the turbojet engine capable of continuously estimating its operational state from available sensor measurements.

4.2 2. Subsystem Health Estimation

Estimate the health state of the following engine subsystems:

- Compressor
- Combustor
- Turbine

Participants should justify the methodology used for health estimation and demonstrate engineering interpretability.

4.3 3. Overall Engine Health Assessment

Develop a unified engine health indicator capable of representing the overall condition of the propulsion system.

4.4 4. Surrogate Modeling

Develop computationally efficient surrogate models that can approximate engine behavior and subsystem states significantly faster than conventional simulation-based approaches.

4.5 5. Performance Prediction

Estimate critical performance parameters such as

- Engine thrust
- Fuel efficiency metrics
- Degradation trajectory

using available measurements and inferred engine states.

4.6 6. Uncertainty Quantification

Provide confidence estimates or uncertainty bounds associated with predictions wherever applicable.

5. Deliverables

The final submission should include the following.

Technical Report

- Methodology
- Feature engineering strategy
- Physics integration approach
- Model architecture
- Validation methodology

Source Code

Complete implementation of the proposed framework.

Digital Twin Dashboard

Interactive visualization demonstrating

- Engine operating conditions
- Compressor health
- Combustor health
- Turbine health
- Overall health index
- Predicted thrust
- Degradation trends
- Prediction confidence

Presentation

A concise summary explaining

- Engineering rationale
- Surrogate modeling strategy
- Health estimation methodology
- Key results and insights

6. Evaluation Criteria

Criterion	Weight
Health Estimation Accuracy	30%
Surrogate Model Performance	20%
Physics Consistency	15%
Generalization Capability	15%
Computational Efficiency	10%
Dashboard and Interpretability	10%

7. Expected Impact

The challenge aims to encourage the development of next-generation aerospace Digital Twin technologies capable of real-time engine monitoring, health assessment, and performance prediction.

Successful solutions may contribute toward predictive maintenance, intelligent propulsion diagnostics, and future autonomous aerospace systems.